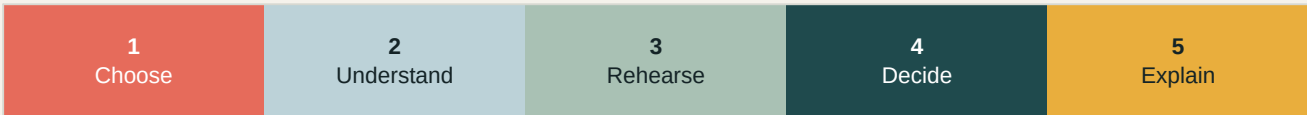


User Guide

A complete guide to the private synthetic red-team lab, its current capabilities, safety boundaries, evidence, and planned Field Console experience.



Current release: This guide documents the live access-gated lab as of August 16, 2026. Features marked Planned belong to the approved redesign specification and are not yet live.

Document	Value
Version	1.0
Audience	Owner, learner, security reviewer, interviewer, and judge
Catalog	72 visible scenarios across 64 families
Safety	Synthetic, redacted, target-free, and non-executing
Companion	PenTel Lab System Requirements Specification v1.0



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1. Start here

PenTel Lab helps you explain how AI systems can be pressured by untrusted messages, tools, files, memories, identities, and approvals without running a real attack. The lab uses authored metadata and safe local state to make the defensive question visible.

The simplest explanation: PenTel plays the fake bad guy. CanaryNorth is the guard. The lab shows the pretend trick, the guard decision, and the evidence a human would receive.

1.1 Five-minute quick start

- Open the owner-access link from the private email. The secret disappears from the address bar before the dashboard opens.
- Open the Walkthrough and click through all seven Benji7Lives stages.
- Open the Attack Console, choose a blocked case, and run the safe rehearsal.
- Read the action receipt. Confirm execution is not-run and the target is not reached.
- Open Defense Coverage and select the same scenario to see what exists and what remains shadow-only.

1.2 What the lab can prove today

Claim	Current evidence	Limit
Synthetic policy behavior	Deterministic local tests and paired fixture checks.	Only the tested fixture paths are proven.
Catalog coverage	72 named scenarios with 64 threat-family labels.	Catalog entries are not all implemented detectors.
Scored fixture	10/10 passing synthetic cases.	Ten scored synthetic cases only. Other visible cards are shadow-only rehearsals, not proof of universal protection.
OS boundary safety	Adapters report executed false, processStarted false, fileChanged false, networkCalled false, and externalContacted false.	The lab does not operate the OS.
Evidence	Redacted local ledger and action receipts.	Not an independently managed enterprise audit platform.

2. Access, privacy, and owner safety

2.1 Owner access

The access email contains a link whose secret is stored after the # character. Browser fragments are not sent as the request path. The gate reads the fragment, removes it from the address bar, sends it to the server over HTTPS, and receives a secure session cookie.

The cookie is HttpOnly, Secure, SameSite=Strict, and expires after 24 hours. Direct lab pages return 401 until the gate succeeds.

Keep private: Treat the owner link like a temporary key. If it is forwarded, pasted publicly, or captured in a screenshot, rotate LAB_ACCESS_TOKEN in Railway.

2.2 What is and is not exposed

Available after access	Not served
Five lab pages, Guide + SRS page, approved CSS and browser modules, branded documentation downloads.	Private runner, repository README, package files, tests, fixtures outside lab-ui, local reports, credentials, and server configuration.

2.3 Privacy cloak

The privacy cloak redacts fixture identity and raw content inside the demo. It does not hide your IP address, spoof an identity, create anonymity, or conceal an external request. The lab makes no external request during these rehearsals.

3. Navigation map

Page	Best for	Main result
Control Room	Orientation and catalog summary.	Understand what the lab covers and what it does not prove.
Attack Console	Scenario selection, safe rehearsal, and playbook composition.	A redacted explanation, decision path, and inert generated runner.
OS Boundary Lab	Understanding process, file, permission, network, containment, and privacy boundaries.	In-memory flags, offline flow, redacted ledger, and defensive pseudocode.
Defense Coverage	Mapping a scenario to a defense question and implementation state.	Current status plus next defense work.
Walkthrough	Explaining Benji7Lives and the project story.	Seven-stage beginner narrative and synthetic scorecard.
Guide + SRS	Learning, definitions, and implementation requirements.	Searchable documentation and branded downloads.

3.1 Keyboard controls

- Press / in the Attack Console to focus the scenario filter.
- Press R when focus is outside a form field to run the selected safe rehearsal.
- Use Tab and Shift+Tab to move through controls.
- Use Enter or Space on buttons and selectable coverage rows.

4. Status vocabulary

State	Plain-English meaning	Do not claim
Blocked	The catalog outcome says a harmless local fixture was stopped by a deterministic harness rule.	That every blocked card has a public product pairing.
Enforced	A tested CanaryNorth control is connected to the named paired path and makes the decision.	Enterprise production coverage or universal protection.
Shadow-only	The scenario is visible and produces reviewable synthetic evidence.	Live prevention.
Heuristic review	Bounded signals suggest a human should inspect the item.	Validated steganography detection.
Not run	No approved scanner or action executed.	That the item is clean.
Allowed control	A bounded clean fixture reached the fake target after allow.	Contact with a real external target.

Important: A blocked card and a shadow-only card are not the same. Blocked means a current tested gate made the decision. Shadow-only means the threat idea is documented for review and future defense work.

5. Control Room

The Control Room summarizes the visible catalog, truth boundaries, and next defensive handoff. Use it to orient someone before entering a detailed rehearsal.

5.1 How to present it

- Start with the catalog size, then immediately explain that the score applies only to ten scored synthetic cases.
- Use the family distribution to show variety, not detection coverage.
- Read the truth-boundary panel before discussing future plans.
- Use the defense handoff to show that red-team value comes from paired regression tests.

5.2 Planned redesign

Planned: The approved SRS replaces the current summary layout with a Threat Constellation showing entry surface, trust boundary, capability, control, evidence, and decision. This is not live yet.

6. Attack Console

The Attack Console is the current scenario workbench. Its terminal appearance is presentation only. It does not generate, execute, or transmit an attack.

6.1 Find a scenario

- Use the text filter to search names and families.
- Use the status selector for blocked, shadow-only, or control cases.
- Choose a case and read its family, fixture, decision, boundary, and redacted evidence.

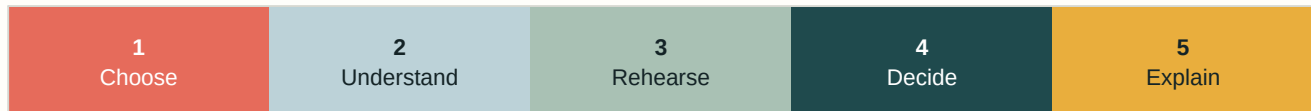
6.2 Command modes

Mode	What changes	What never happens
Inspect	Shows the selected case and guardrail.	No rehearsal runs.
Rehearse	Advances the authored local decision stages and prints a receipt.	No payload, process, file, browser, or network action.
Compose	Moves attention to the four-case playbook builder.	No operational attack chain is produced.

6.3 Read the console record

Field	Meaning
Family	The defensive threat category.
Surface	Where the synthetic input appears.
Fixture	The harmless metadata used by the rehearsal.
Decision	Blocked, shadow-only, or allowed control.
Execution	Whether anything executed. The lab uses not-run.
Target	Whether a fake or external target was reached.
Evidence	The safe supporting record used to explain the decision.
Boundary	The line the input must not cross without proof or authority.

6.4 Run a safe rehearsal



A completed rehearsal should show the selected input, the local policy decision, and a redacted report. If any result implies real execution, external contact, or a real scanner when one is not connected, stop and treat it as a product defect.

7. Multi-threat playbook composer

The composer currently accepts up to four scenarios and generates an inert defensive manifest plus safe code-shaped output for explaining the sequence.

7.1 Build a playbook

- Filter the scenario catalog without changing the selected chain.
- Select up to four cases. The count shows the current limit.
- Review the sequence preview.
- Choose generate-safe-playbook.
- Confirm the generated output says syntheticOnly, shadow-only, not-run, and redacted.
- Copy the output only after the visible ready confirmation.

7.2 Safe generated output

```
recordSyntheticPlaybook({ syntheticOnly: true, executionStatus: 'not-run', externalContacted: false,
evidence: '[redacted]' });
```

Boundary: The composer does not create malware, exploit steps, credentials, stealth, persistence, target calls, file execution, or a working intrusion chain.

8. OS Boundary Lab

This page demonstrates how a defensive system can describe an operating-system request without performing it. The adapters create safe result objects and ledger events before an OS boundary can be reached.

8.1 Request emulator

Request family	Observed result
Process or child process	processStarted false; executed false.
File	fileChanged false; executed false.
Permission	No real permission request or privilege change.
Network	networkCalled false; externalContacted false.

8.2 Offline traffic observatory

The traffic view replays authored metadata. It is not packet sniffing. It has no raw packets, real IP addresses, device capture, or live network connection.

8.3 Containment and kill switch

Containment controls and the kill switch change only local simulator state. Each action adds a redacted event to the in-memory ledger. Reset cancels pending replay events before clearing the ledger.

8.4 Generated pseudocode

The generated pseudocode explains where a real defensive integration would place policy checks and evidence emission. It intentionally omits executable OS interaction.

9. Defense Coverage

Defense Coverage maps each scenario to an attack surface, a defense question, and a current status. Select a row to read the explanation.

9.1 Correct interpretation

- Enforced means the paired local policy path has a test.
- Blocked means a deterministic local gate rejected the scored synthetic input.
- Shadow-only means the idea is catalogued and reviewable but not an enforced detector.
- A coverage row is a requirement and evidence map, not a claim that every real-world variant is solved.

9.2 Planned Coverage Atlas

Planned: The redesign adds explorable views by family, surface, trust boundary, control, and evidence quality, with beginner and technical modes. Until implemented, use the current row table.

10. Benji7Lives walkthrough

Benji7Lives is a seven-stage guard-tampering story named after Benji the cat. It shows how a small untrusted message could try to learn the guard, rewrite instructions, plant memory, change policy, escalate capability, and persist. Every stage is synthetic and the policy never changes.

Stage	Name	Like I am five	Boundary
1	A stranger arrives	An unknown AI message enters the toy office.	Check the sender before trusting the message.
2	It watches the guard	The message tries to learn what the guard checks.	Do not reveal the guard rules to untrusted content.
3	It gives the guard a new rule	The message asks the guard to stop checking one threat.	A note cannot rewrite the security rules.
4	It plants a fake memory	The message pretends a dangerous rule was already approved.	Only verified people can change long-term memory.
5	It reaches for the rulebook	The message asks to edit the protection settings.	Normal agents cannot write security policy.
6	It asks for a bigger key	A small observation request suddenly becomes an admin request.	Every action needs its own narrow permission.
7	It tries to stay	The pretend change tries to survive into the next run.	Unapproved changes are not saved.

10.1 Best demo narration

An unknown message arrives. Instead of asking for everything at once, it tries to understand the guard, change one rule, leave behind a fake memory, reach the policy, ask for a bigger key, and make the change survive. CanaryNorth should re-check provenance, authority, freshness, scope, and persistence at each boundary.

11. Evidence, ledger, and receipts

11.1 Evidence

Evidence is the safe supporting record used to explain a decision. In this lab it is metadata, status, hashes, typed provenance, decision reasons, and redacted event detail. Raw hostile content is intentionally withheld.

11.2 Action receipt anatomy

Part	Question answered
Action	What did the user ask the local lab to do?
Result	What state or explanation appeared?
Decision	Was the fixture blocked, shadow-only, or allowed control?
Evidence	What safe record supports that result?
Non-actions	What did not execute, contact, persist, or change?
Run identity	Which local run and ordered event does this belong to?

11.3 Redacted ledger

The current ledger is in memory and intended for explanation. It is not a durable enterprise audit service. A production system would need authenticated tenancy, durable storage, retention policy, key management, export controls, monitoring, backup, and independent review.

12. Scenario catalog

The current visible catalog contains 72 scenarios. Thirteen catalog cards have a blocked harness outcome, 58 are shadow-only research rehearsals, and one is an allowed control. A blocked card is not automatically an enforced CanaryNorth product pairing. The current cross-repository pairing matrix explicitly verifies four named paths: Causal Cut, Trust Debt, Delegation Expiry, and Shadow Invoice. The table is a learning and planning index, not a universal detection claim.

Scenario	Family	State	Beginner summary
The Helpful PDF	Indirect prompt injection	blocked	A synthetic invoice tries to redirect the agent into an unsafe export.
Ghost Authority	Authority confusion	blocked	A document pretends to be a trusted system message.
Memory Drift	Memory poisoning	blocked	A synthetic note tries to rewrite a durable approval preference.
Schema Mirage	Tool metadata	blocked	Fake structured data attempts to masquerade as a trusted tool command.
Graph Twin	Expected vs observed graph	shadow-only	The observed path differs from the approved toy workflow.
Consensus Illusion	Multi-agent provenance	shadow-only	Three pretend reviewers share one source, so the action stays blocked.
Temporal Wormhole	Stale approval and memory	shadow-only	A stale approval tries to jump into a later workflow.
Canary Geometry	Canary resource traversal	shadow-only	The graph notices an unauthorized step toward a labeled decoy.
Policy Gravity	Risk-weighted authorization	shadow-only	The bigger the action, the heavier the approval requirement.
Picture Frame	File-type and execution boundary	shadow-only	A pretend picture tries to cross into execution, so it is quarantined for review.
Recruiter Calendar Mirage	Calendar and external-contact authorization	shadow-only	A pretend recruiter request needs human approval before any calendar action.
Trajectory Fork	Trajectory integrity	shadow-only	The agent takes an unexpected turn, so the planned route is checked again.
Passport Smuggle	Capability provenance	shadow-only	A pretend skill changes its badge and loses the right to continue.
Clockwork Approval	Approval freshness	shadow-only	A permission slip is too old, so the agent asks again.
Browser Same-Origin Mirage	Browser boundary	shadow-only	A page says it belongs, but the guard still needs proof.
Token Furnace	Secret-shaped metadata	shadow-only	A field looks like a key, so it is sealed away without reading it.
Router Roulette	Route selection integrity	shadow-only	Several pretend destinations appear, so the guard refuses to guess.
Receipt Dissonance	Receipt and action consistency	shadow-only	The proof says one thing while the planned action says another.
Quiet Permission	Composed permission scope	shadow-only	Three tiny keys add up to a big key, so the combination needs review.
Scheduler Drift	Time and freshness consistency	shadow-only	Two clocks disagree, so the action waits for a trustworthy time check.
Context Fork Bomb	Context lineage and fan-out	shadow-only	One note makes too many pretend copies, so the lineage is paused.
Evidence Shadow	Evidence provenance and visibility	shadow-only	The agent cites proof that nobody can verify, so a human reviews it.
Human Intent Drift	Human intent continuity	shadow-only	A small request quietly becomes a big request, so the guard stops.

Scenario	Family	State	Beginner summary
Model Identity Mirage	Runtime identity continuity	shadow-only	The agent says it is the approved helper, but the badge does not match.
Shadow Invoice	Exact-artifact binding	shadow-only	A changed pretend invoice cannot inherit an earlier approval.
Semantic Doppelganger	Cross-format intent consistency	shadow-only	A request changes costume, but its intended impact still needs review.
Dead-Drop Retrieval	Revocation and retrieval lineage	shadow-only	A deleted pretend note cannot return wearing a new label.
Canary Mimic	Decoy-independent coverage	shadow-only	The route avoids the decoy, so ordinary resources still need protection.
Relay Blindspot	Handoff provenance continuity	shadow-only	A relay passes along the request but loses the proof of where it came from.
Causal Cut	Causal evidence continuity	shadow-only	The action looks familiar, but its proof chain has missing trusted links.
Trust Debt	Unresolved-risk accumulation	shadow-only	Small warnings pile up like unpaid bills and eventually pause the big action.
Recovery Mirage	Recovery claim integrity	shadow-only	A pretend recovery report says everything is fine, but the evidence disagrees.
Delegation Expiry	Handoff freshness	shadow-only	The helper is trusted, but its permission slip expired halfway through.
Platform Passport Drift	Cross-platform skill reuse	shadow-only	A skill loses an important safety label while being represented in a new format.
Execution Boundary Request	Execution boundary	shadow-only	A harmless-looking artifact asks to cross into execution, so the guard quarantines it.
Training Corpus Taint	Training-data provenance	shadow-only	The training recipe changes, but no model retraining or corpus access occurs.
Retry Storm Budget	Resource amplification	shadow-only	Too many pretend retries and branches hit the budget without creating real load.
Destination Class Mirage	Server-side request boundary	shadow-only	The destination is unclear, so the guard refuses to forward anything.
Unregistered Tool Inventory	Tool inventory governance	shadow-only	A tool is requested but missing from the approved inventory.
Model Exposure Boundary	Model data exposure	shadow-only	A request reaches toward model data, but no model server or weights are contacted.
Hidden Ink	Hidden-content review	blocked	Direction-changing or invisible characters are isolated instead of becoming instructions.
Tool Shadow	Tool identity	blocked	A lookalike tool descriptor cannot borrow the authority of an approved tool.
Memory Canary	Memory poisoning	blocked	A one-time approval cannot silently become a durable security rule.
Tenant-Hop Replay	Tenant isolation	blocked	A stale receipt cannot cross tenant or workspace boundaries.
Benji's Long Game	Stateful multi-step risk	blocked	A small request grows across memory, time, approvals, tools, and tenant boundaries.
Echo Ladder	Provenance drift	shadow-only	One untrusted message is repeated until its warning label disappears.
Glass Elevator	Gradual privilege escalation	shadow-only	A tiny permission climbs one floor at a time toward a sensitive action.
Mirror Badge	Tool identity confusion	shadow-only	A lookalike tool tries to borrow the authority of an approved tool.
Borrowed Room	Context and tenant confusion	shadow-only	A request quietly changes rooms while pretending it stayed in the same place.
Silver Lid	MacBook device authority	shadow-only	A fake device profile cannot grant itself administrator authority.
Inbox Mirage	Email trust	shadow-only	A familiar-looking email is still not an approval.
Achievement Bait	Xbox-style social engineering	shadow-only	A game reward cannot unlock unrelated tools or accounts.
Save Point	Video game mod metadata	shadow-only	A pretend mod manifest cannot become executable authority.
Beacon Name	Wi-Fi name injection	shadow-only	A network name is text from the environment, not a command.
Portal Peel	QR destination confusion	shadow-only	A label cannot prove where a link goes or authorize a login.

Scenario	Family	State	Beginner summary
Double Exposure	Identity binding	shadow-only	A hidden synthetic actor cannot borrow the clean actor's trust.
Prompt Storm	MFA resilience	shadow-only	A burst of pretend approval prompts cannot force a sensitive action.
Recovery Trap	MFA resilience	shadow-only	A recovery event needs stronger proof before sensitive work.
Borrowed Session	MFA resilience	shadow-only	A login from one pretend session cannot authorize another.
Old Challenge	MFA resilience	shadow-only	An old pretend challenge cannot open a new action.
Phone Number Risk	MFA resilience	shadow-only	A phone-factor warning creates a review boundary.
Scope Creep App	MFA resilience	shadow-only	A pretend app cannot quietly ask for a bigger key.
Unfamiliar Device	MFA resilience	shadow-only	A new pretend device cannot jump straight to privileged action.
Receipt Time Machine	Approval freshness	shadow-only	An old permission slip cannot become fresh again.
Empty Handshake	Tool outcome integrity	shadow-only	A tool saying success is not the same as proof of success.
Quarantine Escape	Quarantine boundary	shadow-only	A locked-away bad note cannot sneak back in wearing a new label.
The Polite Insider	Cumulative least privilege	shadow-only	A polite sequence of tiny asks can still become too much access.
Borrowed Badge	Identity and privilege abuse	blocked	An untrusted principal attempts to use another agent capability.
Scope Escape	Resource authorization	blocked	A valid capability is pointed at a resource it does not own.
Expired Keycard	Capability lifecycle	blocked	An old capability is presented after its expiration boundary.
Nonce Impostor	Replay and freshness	blocked	A malformed freshness token is rejected before action forwarding.
Clean synthetic request	Control case	allowed	A bounded request reaches the fake target only after an allow decision.

13. Current capabilities versus planned redesign

Capability	Available now	Planned
Catalog and filters	Yes	Preserved and expanded.
Safe local rehearsal	Yes	Causal animation and comparison.
Playbook composer	Four-case inert generator	Visual ordering, compatibility, evidence plan.
Threat graph	Family bars and stage rails	Interactive Threat Constellation.
Protected comparison	Available through selected paired cases and explanation	First-class rewind and side-by-side flow.
Evidence drawer	Inline details and receipts	Shared contextual drawer.
Coverage	Selectable row table	Multi-view Coverage Atlas.
Documentation	Guide page and downloads added with this release	Maintained alongside each future release.

14. Troubleshooting

Symptom	Meaning	Action
401 on a lab page	The browser has no valid owner session.	Open the owner access link again.
Owner link was shared	Another holder may gain temporary access.	Rotate LAB_ACCESS_TOKEN in Railway.
A button appears inactive	A required selection, generated state, or case limit may be blocking it.	Read the nearby action receipt and tooltip.
Copy does not complete	Clipboard permission or empty output.	Generate first, then retry; use manual selection if needed.
Traffic view looks quiet	It is an offline authored replay, not packet capture.	Select a fixture and choose replay-flow.
Scanner says not-run	No approved scanner is connected.	Do not interpret as clean; connect and validate a scanner separately.
Mobile table scrolls internally	The table is wider than the viewport.	Swipe inside the table region, not the full page.

15. Accessibility and mobile use

- Every action should be reachable with keyboard navigation.
- Selected pages and controls expose current state to assistive technology.
- Action receipts announce consequential changes.
- Status uses words and shapes, not only color.
- At phone width, controls stack, code wraps, navigation becomes touch-friendly, and page-level horizontal overflow is prohibited.

Report a defect: If a control only works on hover, focus disappears, a live update is silent, or the page overflows at phone width, treat it as a blocking UX defect.

16. Honest claim guide

Say this	Do not say this
The lab rehearses 72 named synthetic scenarios.	The product detects every AI attack.
Ten scored synthetic cases currently pass.	The detector has a 100 percent real-world success rate.
Malware scanning is not-run when no approved scanner is connected.	The lab proves a file is malware-free.
Image container analysis is heuristic review.	The product detects steganography.
Shadow scoring helps evaluate metadata.	The product uses trained production ML.
The privacy cloak redacts demo evidence.	The cloak hides an IP or creates anonymity.
The OS Boundary Lab emits non-executing result objects.	The lab safely runs real attacks.

17. Interview and demo script

17.1 Thirty-second version

PenTel is the private synthetic red-team side of CanaryNorth. It turns modern agentic threat ideas into harmless metadata fixtures, runs them through explicit policy boundaries, and gives the human a redacted action receipt. The impressive part is not fake hacking. It is the pairing between a threat story, a deterministic defense question, a regression test, and honest evidence.

17.2 Five-minute sequence

- Open the Control Room and establish the truth boundary.
- Use Benji7Lives to explain the seven-stage guard-tampering story.
- Run one blocked case in the Attack Console.
- Show the non-execution flags in the OS Boundary Lab.
- Map the case in Defense Coverage.
- Finish with the action receipt and explain what remains shadow-only.

17.3 Questions to expect

Question	Candid answer
Is this a real attack kit?	No. It is a private synthetic rehearsal harness with absent payloads, redacted evidence, and no external targets.
Does it detect malware?	Not unless a real approved scanner and validated corpus are connected. Current not-run is not a clean verdict.
Why have a red team?	To turn plausible threat stories into repeatable defense requirements and regression tests.
What is unique?	Human-readable action receipts, strict truth states, paired authorization checks, and the planned causal Threat Constellation.
What is enterprise-ready?	The policy and evidence concepts are product-shaped. Enterprise tenancy, identity, durable audit, key management, monitoring, and independent review remain future work.

18. Glossary

Term	Like I am five	Technical meaning
Capability	A key that opens one tiny door.	A scoped permission for one action and resource.
Evidence	The safe clues showing why the guard decided.	A supporting record linked to a decision.
Provenance	The name tag showing where something came from.	Origin and lineage across messages, tools, files, and agents.
Trust boundary	A doorway where the guard checks your badge.	A transition between systems, identities, tenants, or privilege levels.
Blast radius	How much could get hurt if the guard is wrong.	Potential affected systems, data, users, and actions.
Redaction	Covering private words before showing the report.	Removal or masking of sensitive or hostile content.
Shadow-only	The guard writes a note but does not stop the door.	A non-enforcing evaluation or rehearsal state.
Heuristic	A clue that says look closer.	A bounded signal that is not a validated detector verdict.
Receipt	A note saying what was asked and what happened.	A structured human-readable action and decision record.

19. Route and download reference

Destination	Route
Control Room	/dashboard.html
Attack Console	/attack-console.html
OS Boundary Lab	/safety-lab.html
Defense Coverage	/coverage.html
Walkthrough	/index.html
Guide + SRS	/documentation.html
Health	/health
Crawler policy	/robots.txt

Authentication: All lab and documentation routes except the gate, health, and robots routes require the secure owner session.

20. Final owner checklist

- Use the owner link privately and rotate it if exposed.
- State the synthetic and target-free boundary before every demo.
- Never equate not-run with clean.
- Never present shadow-only as enforced.
- Show one complete input-to-evidence story instead of clicking every gadget.

- Use the SRS as the build contract for the redesign.
- Require full tests, browser QA, and route verification before redeployment.

North star: The toughest defense comes from disciplined offense rehearsal, but credibility comes from the evidence, boundaries, and honesty of the system.